

115.109

Introduction to Business Analytics

Topic 4: Customer Trends Variation



Dr Andrew Murphy
(Please contact me via Keeping in
Touch in Stream)

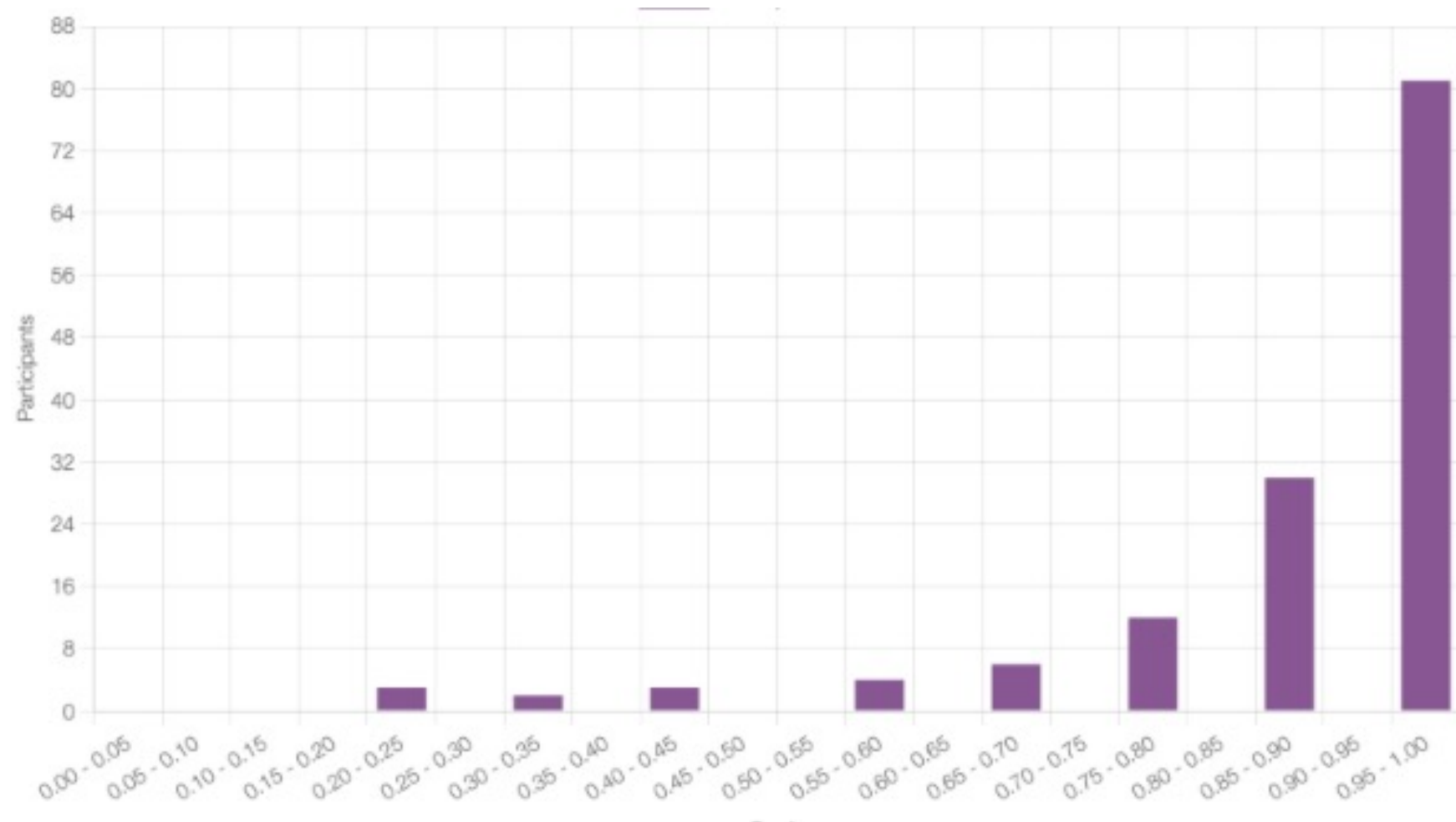
Week (starting)	Business Problem	Data & Analysis	Chapter (Black et al)	Assess ments
1 (13/7)	Understanding Data: discounts & margin	Calculating data in Excel; GST, discounts and Margins	1	Lab1
2 (20/7)	Operational Monitoring: Dashboards	Tabulating & Graphing	2	Lab2
3 (27/7)	Industry trends	Statistics NZ: retail survey Descriptive stats: averages	3	Lab3
4 (3/8)	Customer trends	Statistics NZ: census Descriptive stats: variation	3	Lab4
5 (10/8)	Transaction data: product sales	Transaction data sources. Sampling	7	Lab5
6 (17/8)	Loyalty cards	Loyalty card schemes. Probability; proportions, χ^2	4, 12	Lab6 Test 1
Mid	Semester	Study	Period	
7 (7/9)	Segmentation	Statistics NZ: census. Normal distribution; Means	6, 8	Lab7
8 (14/9)	Determining difference (1)	Means; ANOVA	9, 10, 11	Lab8
9 (21/9)	Associating and Predicting	Explaining relationships: Correlation & regression	13	Lab9
10 (28/9)	Determining Difference (2)	Explaining relationships (continued)	14	-
11 (5/10)	Trends over time	Time series, Indices Unstructured data: social media	15	Lab11 Test 2
12 (12/10)	Corporate reporting	Finance Dashboards reporting		Report

Assessments Schedule

[illegible]

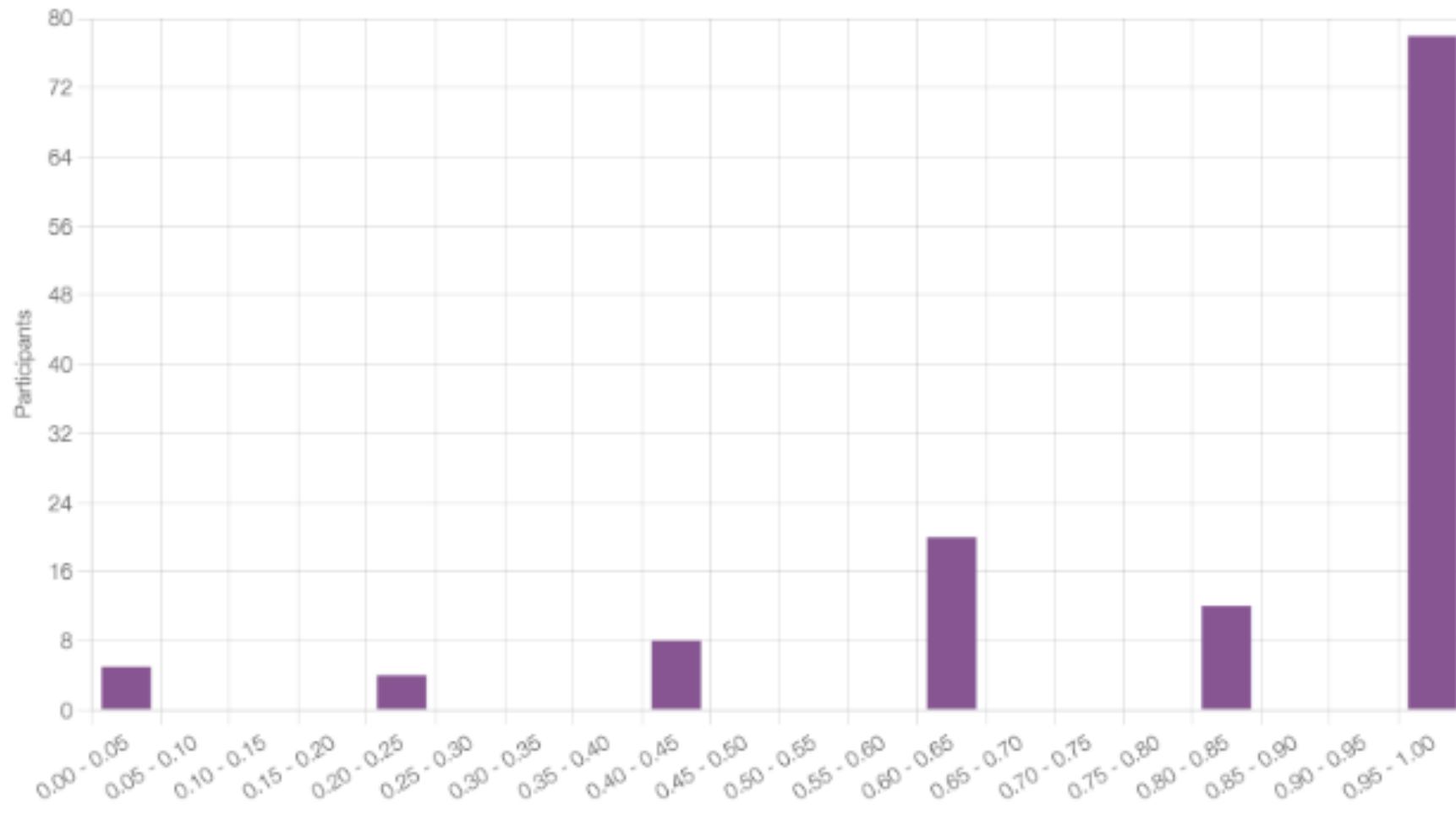
Labs 1&2: as of 4/8

- 139 submitted lab file
- 139 completed Lab 1 Red Box Quiz ($\bar{x}$: 0.89)



Labs 1&2: as of 4/8

- 139 submitted lab file
- 139 completed Lab 1 Red Box Quiz ($\bar{x}$: 0.89)
- 127 completed Lab 2 Red Box Quiz ($\bar{x}$: 0.82)



Group Information Survey

Contents

- ↓ [Assessment Overview](#)
- ↓ [Assessment Planner](#)
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- ↓ [Missing AI Declarations?](#)
- ↓ [Labs 1 & 2](#)
- ↓ [Labs 3 & 4](#)
- ↓ [Assessment 2 & 3 information: Online Tests](#)
- ↓ [Optional Reviews](#)
- ↓ [Assessment 4 information](#)
- ↓ [Analytical Report](#)
- ↓ [Report Groups](#)

21 groups
so far



QUESTIONNAIRE

Established group identifier: closes 10 August



Enter the Student IDs of ALL of the group members (either 2 or 3 members per group) IF you have already formed a group, or know of other students in the class you would like to form a group with.

If you have not yet formed a group, I will allocate you - complete the [Group Information Survey](#) to assist this.

40 as of 4/8



QUESTIONNAIRE

Group Information Survey: closes 10 August



If you have not already formed a group, please complete this survey by Monday August 10

This survey will help me generate teams of similar learners for Assignment 4 (group report). All information in the survey is confidential and is used solely for the purpose of organising teams for this assessment.

Teams created using the survey information tend to be better organised than teams randomly assigned.

Items with black * in the question will be shared publicly with your teammates (name, email and city only). The red asterisk next to the question number indicates a required question. This information is not shared.



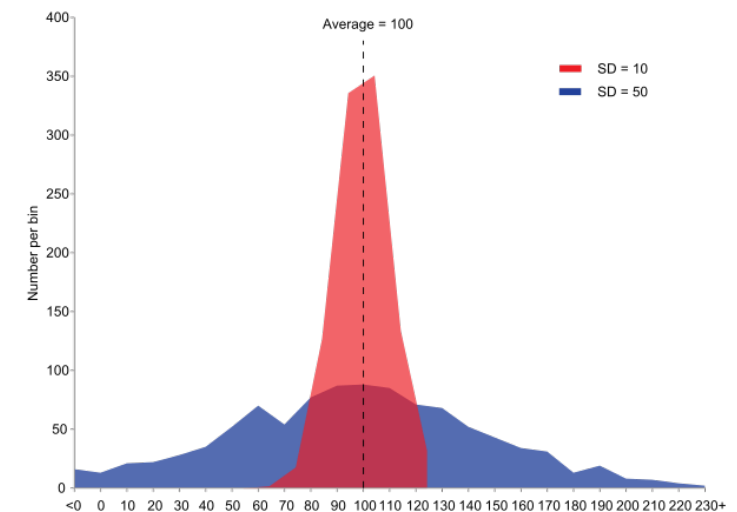
Assessment: Test 1 (10%)

- Available online from evening of Friday 14th August (5pm) to Friday 21st August (5pm)
- Available for 60 minutes from time first started
- Can be started any time during the period, but must be completed in one session – cannot pause and restart later (beyond the 60 minutes)
- Open book (Black et al. would be OK to have in front of you)
- Weeks 1-5 included
- There won't be questions on material in the text not covered in lectures or labs, but there may be questions on material in the lectures/labs that is not specifically in the text(s).
- Questions will be multiple choice.
- You may be asked to do some calculations, so have access to a calculator (Excel, calculator App on computer or phone)
- Multiple choice answers will be available once the test window has closed. Grades will show in the Stream grade book.



Dispersion

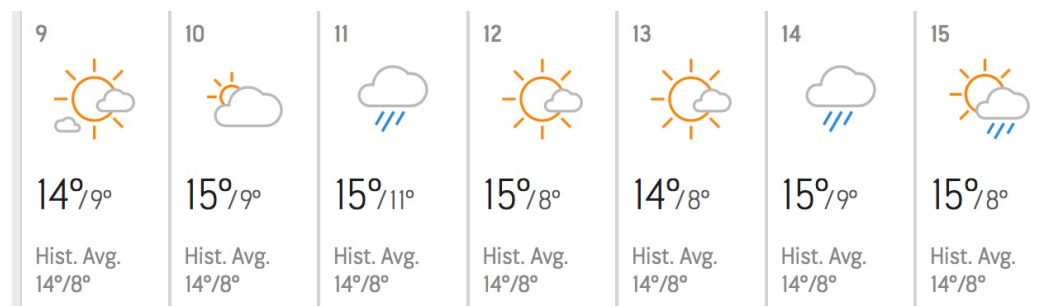
- A measure of central tendency in itself is not sufficient to describe a set of data adequately
- A measure of *variation* (or *dispersion*) of the data is usually required
- This measure gives an indication of the internal *spread* of the data—that is, the extent to which data items vary from one another or from a central point
eg Range
- Some reasons for requiring a measure of dispersion of a set of data:
 - As an indication of the *suitability* of the average value
 - To identify outliers
 - To assist in *controlling* unwanted variation



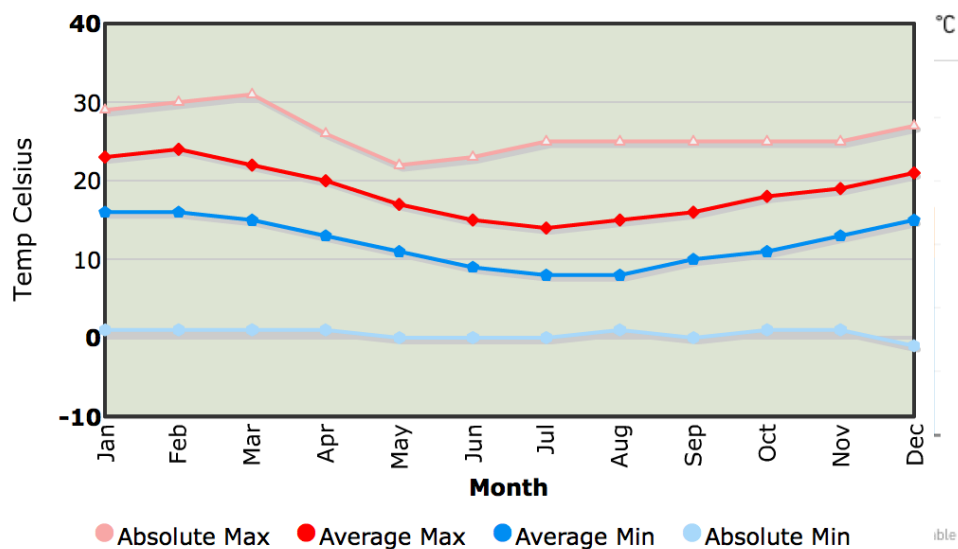


Range: Maximum – Minimum

- Temperature fluctuations on a given day

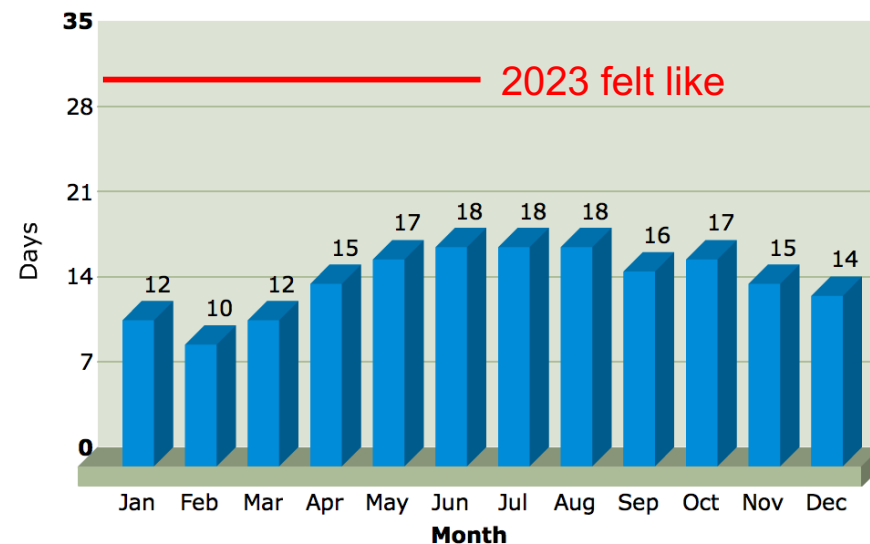


Temperatures: Averages and Extremes



<http://www.myweather2.com/City-Town/New-Zealand/Auckland/climate-profile.aspx>

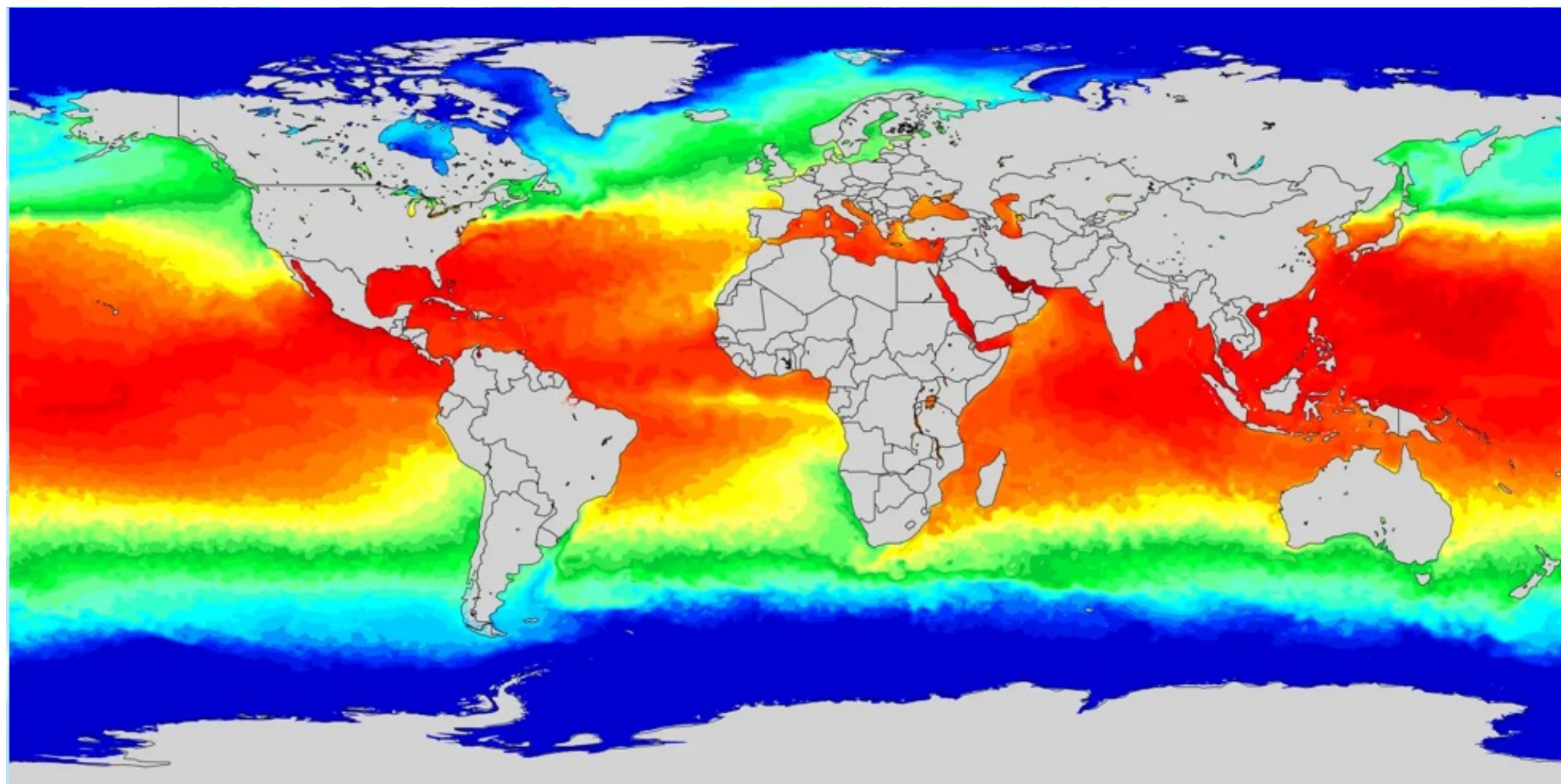
Number of Rain/Drizzle days in month





Range: Maximum – Minimum

World Sea Temperatures 26th Jul 2026



°C 0 1 3 5 7 9 11 13 15 17 19 21 24 27 30 35

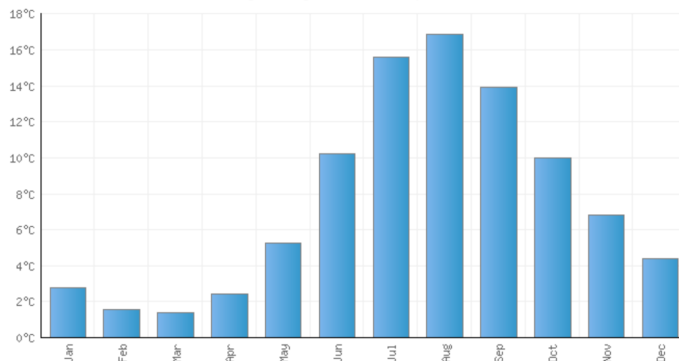
<https://www.seatemperature.org>



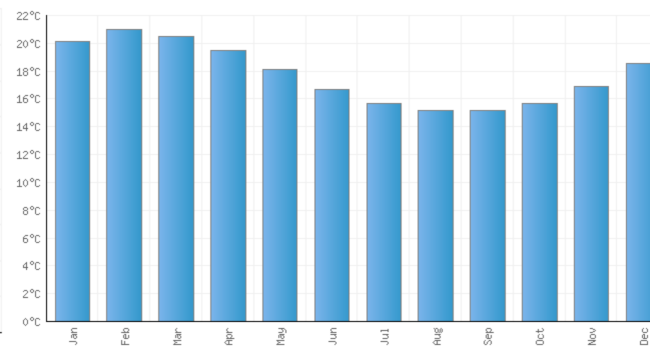
Range: Maximum – Minimum



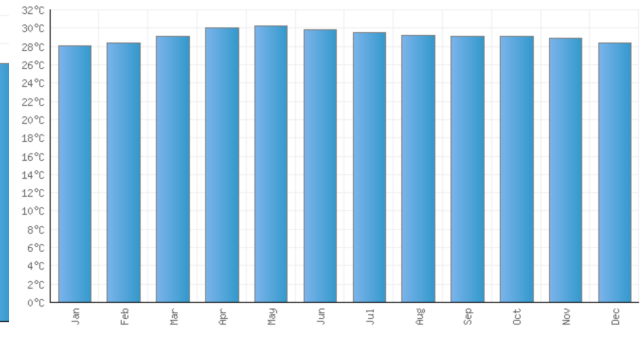
Average monthly Helsinki sea temperatures (°C)



Average monthly Auckland sea temperatures (°C)



Average monthly Ko Samui sea temperatures (°C)





Auckland Sea Temperature

Auckland Water Temperature

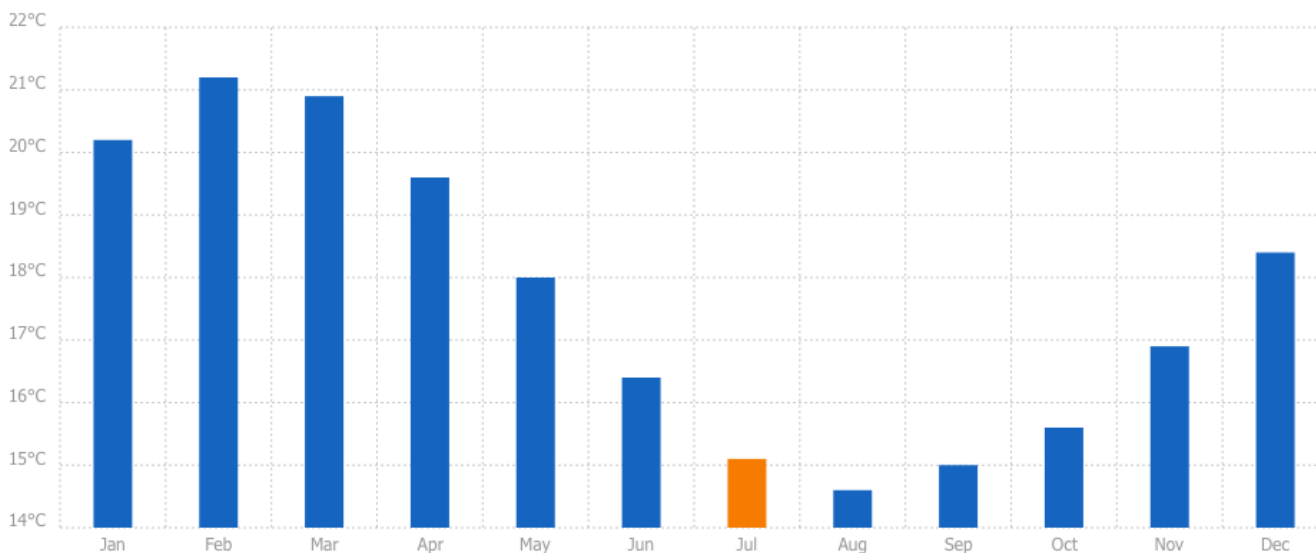


The water in Auckland today will feel **cool**

The water will feel a bit chilly in just bathers / swimming costume. If you plan on wearing a wetsuit 4/3mm should be thick enough.

Monthly average water temperatures

The graph below shows the range of monthly Auckland water temperature derived from many years of historical sea surface temperature data.



	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Min °C	17.8	19.7	19.8	18.3	17.2	15.5	14.1	13.8	13.8	13.9	15.2	17
Avg °C	20.2	21.2	20.9	19.6	18	16.4	15.1	14.6	15	15.6	16.9	18.4
Max °C	22.7	22.8	22	20.9	18.9	17.4	16.2	15.5	16.2	17.3	18.6	19.9

Range: Movement of share prices / currencies (high-low-close)



<https://finance.yahoo.com/chart/AAPL>

Apple Inc. (AAPL)

333.02 +11.36 (+3.53%)

336.01 +2.99 (+0.90%)

☆ Follow

NasdaqGS - BOATS Real Time Price • USD

At close: July 24 at 4:00:01 PM EDT

Overnight: 9:53:38 PM EDT ⓘ

Mounta

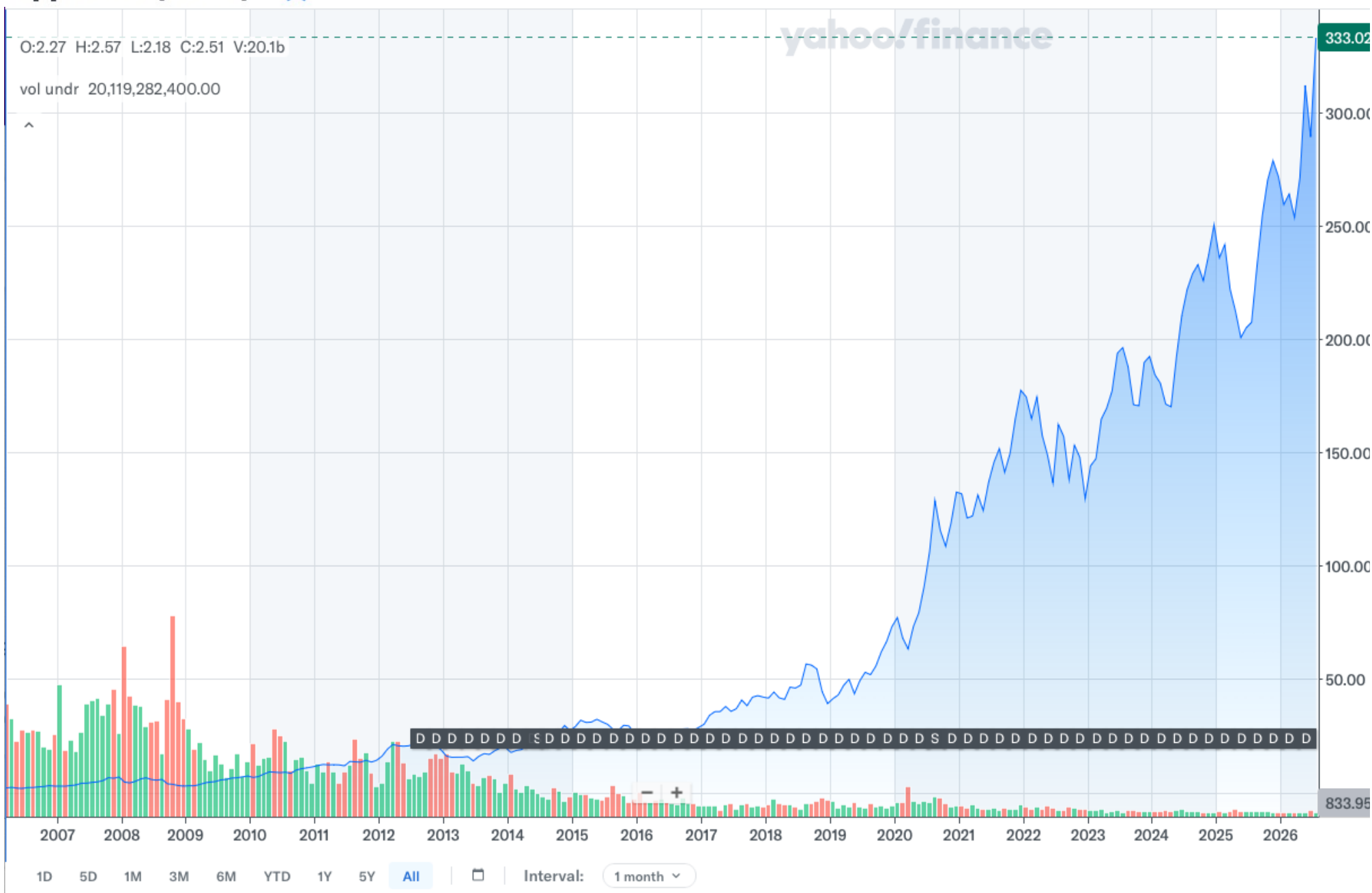




Range: Movement of share prices / currencies (high-low-close)

<https://finance.yahoo.com/chart/AAPL>

Apple Inc. (AAPL) ☆



SpaceX IPO

<https://finance.yahoo.com/chart/SPCX>



Tesla

<https://finance.yahoo.com/chart/TSLA>

Tesla, Inc. (TSLA)

313.03 -6.66 (-2.08%)

315.29 +2.26 (+0.72%)

☆ Follow

Canc

NasdaqGS - BOATS Real Time Price • USD

At close: July 24 at 4:00:01 PM EDT

Overnight: 11:43:54 PM EDT ⓘ

O:320.72 H:322.96 L:306.51 C:313.03 V:62.6m

vol undr 62,575,300.00

yahoo!finance





NZD to 1 British Pound



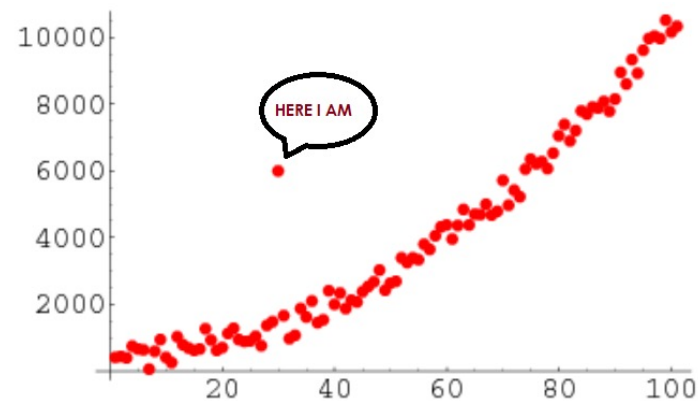


Range: Examples

- Temperature fluctuations on a given day
 - Movement of share prices / currencies (high-low-close)
 - Acceptable systolic and diastolic blood pressures, cholesterol levels
 - Data quality checking: scales used correctly?
-
- Range is considered primitive as it considers only the extreme values, which may not be useful indicators of the bulk of the sample/population

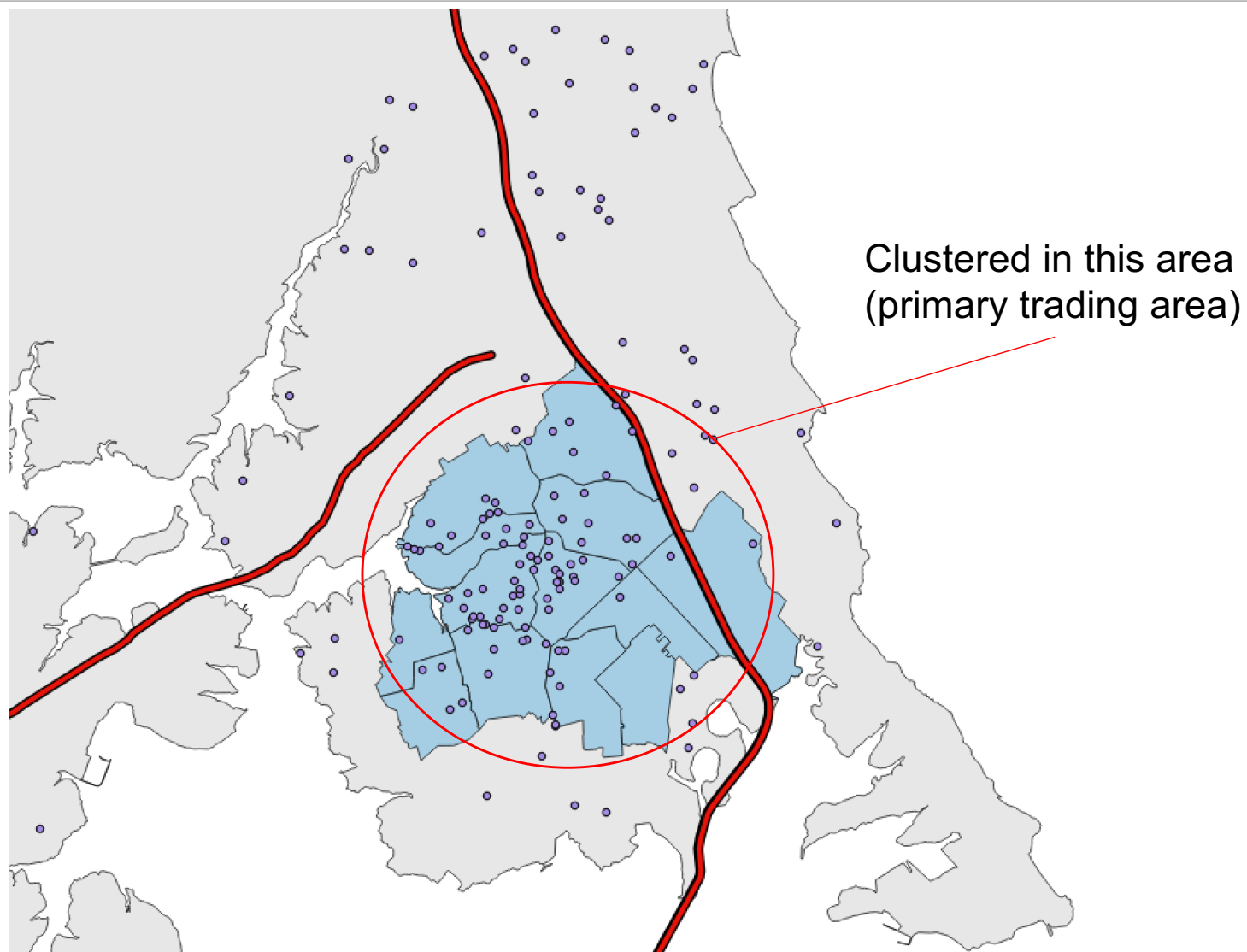
Extreme values (*outliers*), may result from errors of measurement or coding

- *Outliers* are values that are inconsistent with the rest of the data



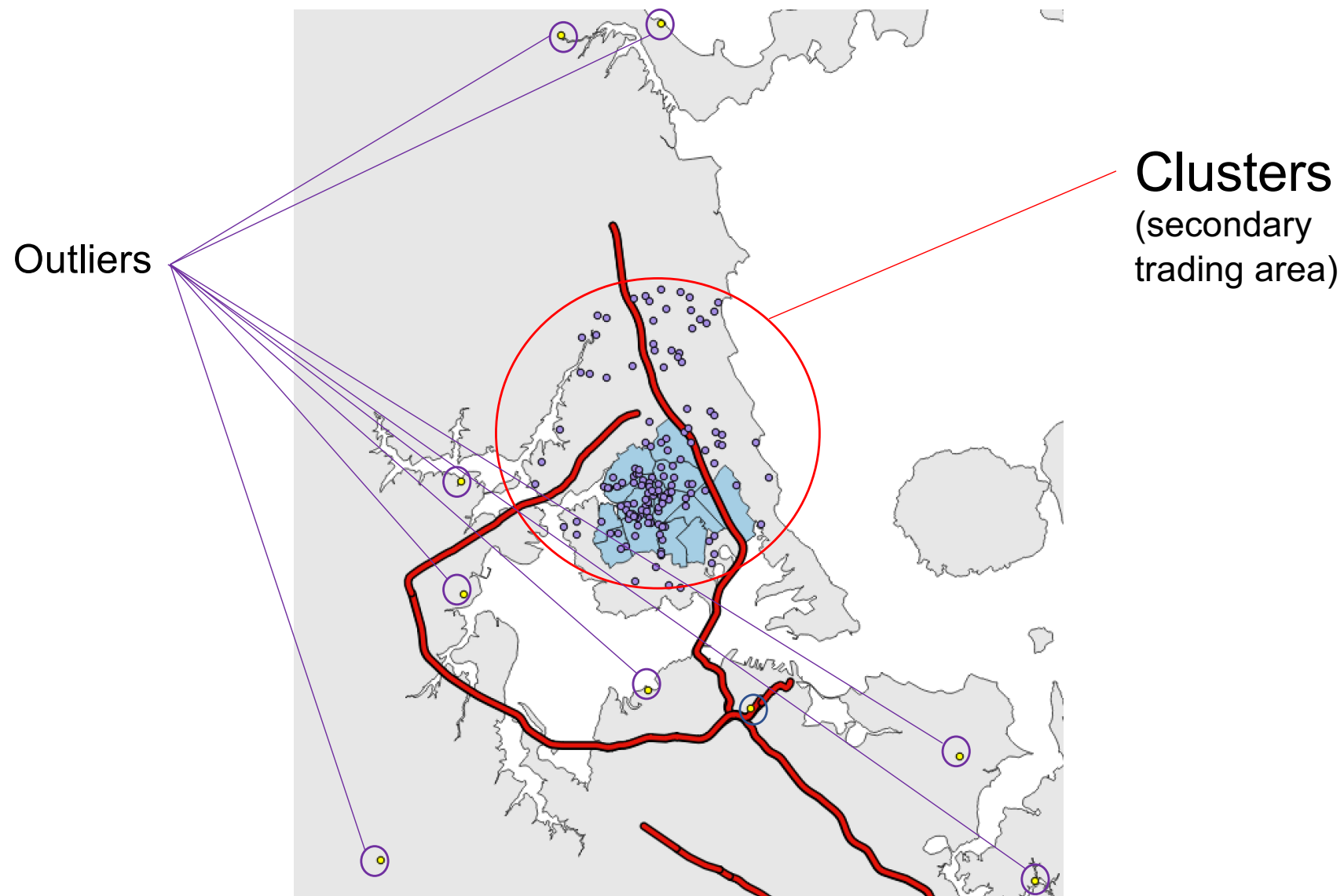


Customer Locations: Glenfield Countdowns





Outliers





Andrew Arnold • 1st

Country Manager - New Zealand at Shopper Intellig...

1w • 🌐



Catering to the time poor shopper who is occasion or mission driven is so important, so it was great to see this in the flesh at New World Long Bay earlier today. Great location (right at the front of the store), great range, flexibility in what you can choose and the callout is on point - "looking for a hassle free dinner" speaks exactly to the mindset this type of shopper is in.

How big is this opportunity I hear you ask? Consider this - 40% of all shoppers are buying with a specific meal occasion in mind, and 30% of all shoppers are buying to consume or use today. To me that is a significant amount of unlocked potential to provide a solution for shoppers who want something right here, right now.

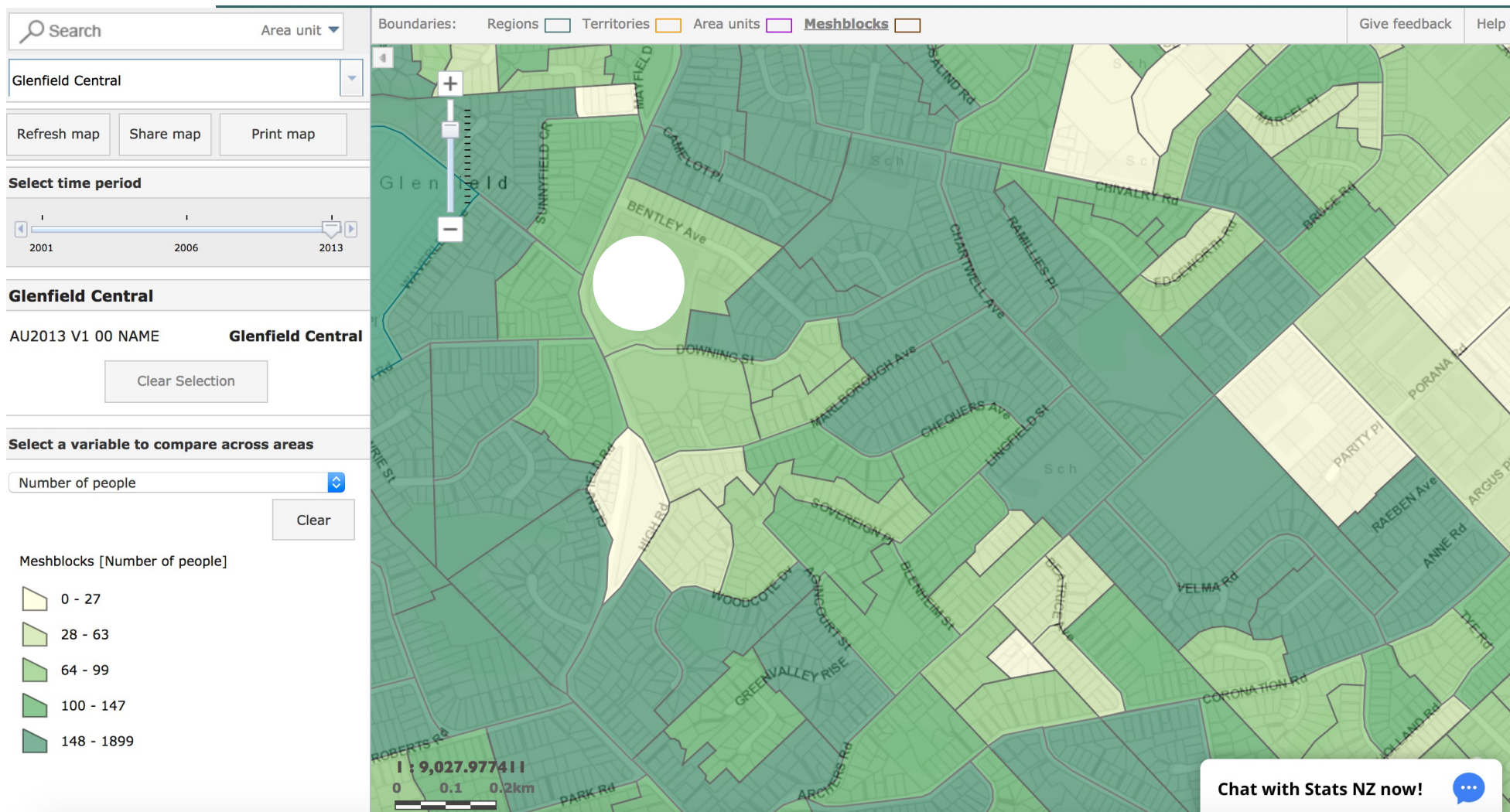
And did I mention that the cold drinks fridge is two steps to the right of this display, and the barista is right behind you? All bases covered!

Shopper Intelligence ANZ #occasion #mission #whatsfordinner



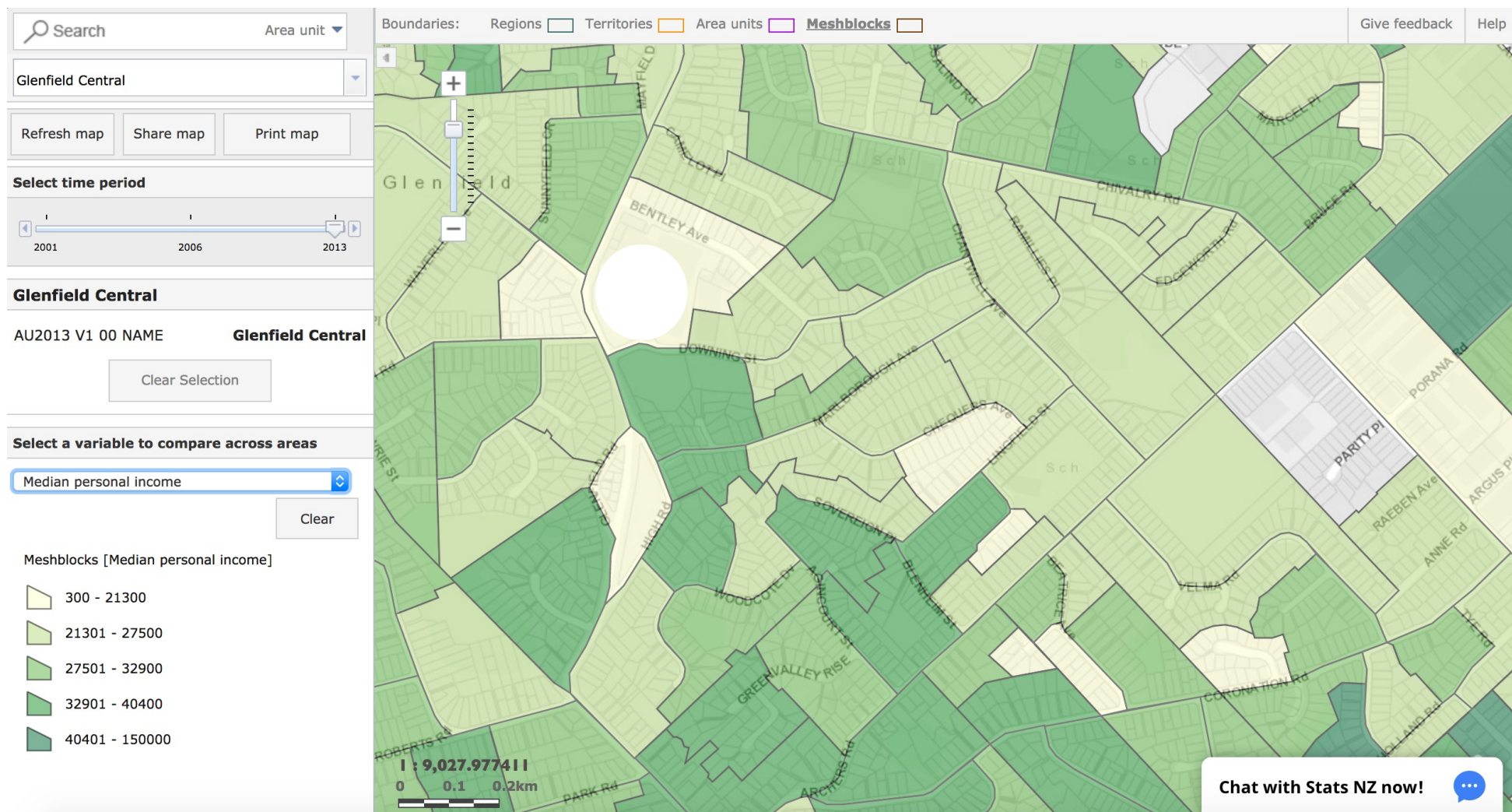


NZ Census: Glenfield



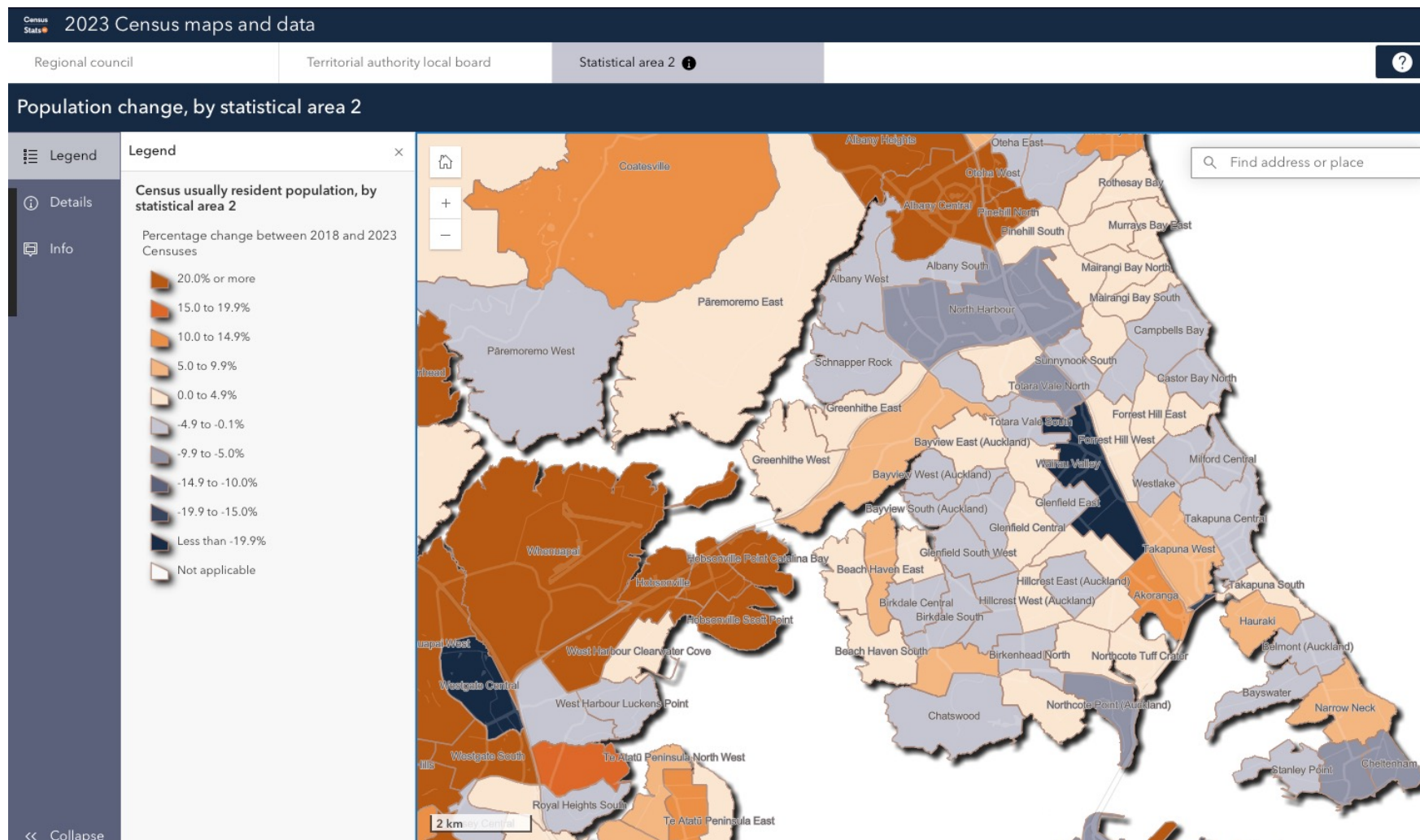


NZ Census: Glenfield





NZ Census 2023: Population change



Lab 4: Sustainability Summary Statistics



115.109 Topic 4 Lab Worksheet: Descriptive Statistics

Due date: August 17

Version: 1 (20.07)

Before you proceed: please complete the Sustainability Survey yourself, here ==> https://masseybusiness.asia.qualtrics.com/jfe/form/SV_d0Vcl6mFNp9tUCW

Task 1: Determine the Demographics of the sample in the SustainabilityData sheet (next tab), as follows

Rationale:
This lab comp "Sustainability omnibus surv In a future we class survey fi

Task 1 looks a

In Task 2 we l compare thes

1.1 Gender (column C)

	1 (Male)	2 (Female)	3 (Gender diverse)	Total
# Respondents				
% Respondents				

To construct PivotTable:

For Gender: click on column C of the SustainabilityData tab > Insert > Pivot Table (I started the selection from C4 in the example, however, you can use the entire column. In the pivot table you can later deselect the values you do r

Drag field name "Gender" to the Rows box, and again drag field name Gender to the Values box (which changes to "Count of Gender"). Your summary table should now appear.

Copy the frequency counts (Paste Special > Values to the table above, then calculate % of total in cell C19, and copy and Paste Special > Formulas across, to determine % Respondents

Note: you can select multiple columns when you create PivotTables, then swap data in and out of the table. Better yet, duplicate (copy/paste) the PivotTable, and change the data source

Hint: to get a count of Male, you could use the function COUNTIF, like we did in Lab 1. Eg for the data range B5 to B1046 : =countif(B5:E1016,1) {where 1='Male', 2='Female', 3='Gender diverse' - please note this formula only works within the SustainabilityData sheet, not from here}

Or create a single Pivot Table (instructions below)

1.2 Age category (column D)

	7 (<18)	1 (18-24)	2 (25-34)	3 (35-44)	4 (45-54)	5 (55-64)	6 (65+)	Total
# Respondents	1							1042
% Respondents	0.1							

1.3 Age category column chart and Histogram

Using cells C17:I18 above, create a column chart of the age categories below. In the SustainabilityData sheet, create a Histogram of the Age column, and cut and pate that chart here. Which is a better representation of the data?

If your Column shows two series (a taller series showing C18:I18 and a shorter series showing C19:I19): click on any one of the shorter bars, and press the delete key to remove (and then remove the legend)

The Histogram defaults to some pretty ugly class intervals. Right-click on any one of the bars to bring up the Format Data Series panel, and change Bins from Auto to Bin width = 1 (and adjust the underflow and overflow as per be

Pivot Table Analyse

Insert Slicer Insert Timeline Filter Connections

Design

Refresh Change Data Source

1.4 Income (column CE) (via PivotTable)

Click on column CE of the SustainabilityData tab > Insert > Pivot Table. Drag field name "Income" to the Rows box, and again drag field name Income to the Values box (which changes to "Count of Income")

Paste Special > Values to row 42 below, then calculate the % values below

	1: Up to \$30,000	2: \$30,001 - \$50,000	3: \$50,001 - \$70,000	4: \$70,001 - \$100,000	5: \$100,001 - \$130,000	6: Over \$130,000	7: Would rather not say	Total
# Respondents								
% Respondents								

Andrew Murphy:
We could remove the three respondents that took "too long" to complete the survey, but we don't know why - perhaps they stepped away for a cup of coffee and forgot to submit?

If interested, create two Boxplots: one with all the data, the second omitting the three with completion times > 50 minutes

What about those that were "too fast"? {Perceptive have already screened these out.}

1.5 Completion time Boxplot

Create a boxplot of the completion time - using the data in column B of the SustainabilityData sheet. (We have more data here than in Lab 1 Task 3, but v

Select column B rows 5 to 1046, then go to the 'Statistical' chart group and choose Box and Whisker

Give the chart a proper title and remove the X-axis data labelling (which is often pretty meaningless in Excel's version of the chart), and add an

Place chart here: (resize chart to fit)

Optional boxplot: Truncated data (ignoring those over 50 minutes)

[illegible]

Lab 4: Sustainability Summary Statistics



84 **Task 2: Calculate sample size (N), Mean (xbar) and Std Deviation for the data in SustainabilityData columns BD and BE (To what extent do you agree with the following statements, using scale 1:**

85 **BD: Retailers should be required to take back packaging they sell product in (including cardboard, bubble, wrap or filler, polystyrene...)**

86 **BE: There should be a bottle deposit scheme, with customers reimbursed for taking back milk, juice and water bottles to the retailer or a recycle station**

Hint: If you have constructed your statistics in the S ('Paste Special' Values [to remove the formulas] and
Alternatively you can complete the formulas here, I

89	Agree/disagree Questions		Take Back Packaging (BD)				Bottle Deposit Scheme (BE)					
		Scale Responses (x)	Frequency count (f)	fx	Residuals: $(x-\bar{x})$	Residuals squared: $(x-\bar{x})^2$	$f * \text{sqd Residuals}$ $f(x-\bar{x})^2$	Frequency count (f)	fx	Residuals: $(x-\bar{x})$	Residuals squared: $(x-\bar{x})^2$	$f * \text{sqd Residuals}$ $f(x-\bar{x})^2$
90			1									
91			2									
92			3									
93			4									
94			5									
95			Sum									
96			$\bar{x}$ (mean)									
97			s (std dev)									
98												
99												
100	Short label		(sample size) N	Mean	Standard Deviation							
101		Return Packaging	1042									
102			=count()	=average()	=stdev()							

	A	AW	AX	AY	AZ	BA	BB	BC	BD	BE	
1	Variable Game to mind when you think of sustain Sustainable intentions and behaviour										So
2	Question	ThirdBrand	FourthBrand	FifthBrand	How often do	How likely are	I am willing to	I believe it is a	Retailers shou	There should	Con
	Variable coding:	(Free text)	(Free text)	(Free text)	1: Not at all 5: Very often 6: Don't know	1: Not likely 5: Very likely	1: Strongly disagree 3: Neither disagree nor agree 5: Strongly agree	1: Strongly disagree 3: Neither disagree nor agree 5: Strongly agree	1: Strongly disagree 3: Neither disagree nor agree 5: Strongly agree	1: Strongly disagree 3: Neither disagree nor agree 5: Strongly agree	1: N 5: Ex Imp
3											
4	Variable	ThirdBrand	FourthBrand	FifthBrand	HowOftenB	LikelyPayM	WillingPay	Acceptable	Return	Deposit	Sol
1044	P1040	Glassons	Ballantyne	Mercury	uy	ore	More	PayMore	Packaging	scheme	soc
1045	P1041		Eco warehouse								
1046	P1042	Pagani									
1047											
1048	Male				1	33	71	75	6	39	21
1049	Female:				2	53	114	124	10	69	28
1050	Diverse:				3	240	381	383		256	162
1051					4	261	383	370	42	377	324
1052	Max				5	154	93	90	11	301	507
1053	Min				6	301					
1054	>50	=count(Ay5:Ay1046)			N	741				1042	1042
1055		=average()			Mean	4.298464					
1056		=stdev()			Standard deviation						
1057						3.607287					

Lab 4: Sustainability Summary Statistics

105	Bonus Task 3: Qualitative Analysis of Brands																		
106	PivotTables not only analyse quantitative data - they can tabulate qualitative data too.																		
107	1) Construct a PivotTable of FirstBrand ... FifthBrand (columns AT:AX) for SustainabilityData), and paste results below. READ ALL THESE INSTRUCTIONS!																		
108	<i>It would be tempting to put all the Brands columns into the Values box at once, but unfortunately that doesn't work. Each Brand needs its own separate PivotTable</i>																		
109	<i>Ideally we would be coding these entries: converting store names to a number (making allowances for alternative spellings of the same company) and counting the codes</i>																		
110	<i>[eg 'Warehouse' and 'The Warehouse'; 'EcoStore' and 'Ecostore' and 'Eco Store' and 'Eco store' ...].</i>																		
111	<i>Doing so usually involves two people doing the coding independently, and cross-checking. For example: is "count" short for Countdown, Country Road, or something else?</i>																		
112	<i>Is Eco Shop the same as ecostore? (Which is the correct capitalisation, even though no respondent put it that way!) What about Eco?</i>																		
113	<i>What do we do with all the "I don't know" or "Can't think" or "None" or ... (there are many variations on this, because a response (of any kind) was required for FirstBrand.</i>																		
114	<i>To simplify this a bit, I have created a new tab (All Brand Mentions) which reproduces the FirstBrand column, removing all non-brand mentions that I could easily identify.</i>																		
115	<i>I then combined in a second column all the First, Second....FifthBrand mentions into one continuous column, sorted alphabetically.</i>																		
116	<i>As a result there are 708 FirstBrand mentions (some of which might still be nonsense, and many of which can be collapsed due to variations in spelling); and 1729 mentions of any brand in any ranking</i>																		
117	<i>Use the All Brand Mentions tab to create your Pivot Tables.</i>																		
118	2) Which five brands are most frequently top-of-mind? (ie, first mentioned)																		
119	<i>Create a FirstMentions PivotTable, and sort-descending (right-click any result row eg Count of FirstMentions for Allbirds) and Sort > Sort Largest to Smallest. Paste the results below under</i>																		
120	<i>Bear in mind that the ranking might change once you combine the multiple spellings, name variations, and sub-brands</i>																		
121	3) Which five brands are the most mentioned overall (any rank)?																		
122	<i>Create an AllMentions PivotTable, and sort-descending (right-click any result row eg Count of AllMentions for Allbirds) and Sort > Sort Largest to Smallest. Paste the results below under 'A</i>																		
123	NB: To navigate any large data file, you can freeze the first row, so the headings stay put as you scroll down. Click the View menu > click Freeze Top Row																		
124	NB: to freeze multiple rows (and/or columns), click in a cell BENEATH/RIGHT OF where you want to freeze, and click 'Freeze Panes'. (That is what I did to the SustainabilityData file, where th																		
125																			
126	Top-of-Mind: Top 5				Top 5: Any ranking														
127	Rank	Brand/Company		Frequency		Brand/Compan		Frequency											
128	1																		
129	2																		
130	3																		
131	4																		
132	5																		
133																			
134	<<End of Lab 4>>																		